Quantifying a social problem - The dutch housing crisis

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Identification of the social problem

Describe the Social Problem

The Netherlands has been facing a significant housing shortage since the 1990s. The demand for affordable and suitable housing continues to exceed the available supply, making it increasingly difficult for many people to find a home. This issue affects a wide range of groups, including young adults, students, low- and middle-income families, and first-time homebuyers. This topic is relevant because it raises urgent questions for young adults and students about their future housing prospects, such as whether affordable housing will be accessible after graduation and where they will be able to live. This makes the problem both relevant and urgent. Research suggests that the Dutch housing shortage is partly linked to the inefficient use of existing housing stock, as many households occupy more living space than they actually need. As a result, a mismatch exists between the available housing and the needs of different population groups, making it more difficult for starters and smaller households to find suitable homes (Hochstenbach, 2026). Further causes of the shortage include factors like population increase, urbanization, low capacity for housing construction, slow housing construction process, and lack of adequate housing development in the past few years. The Dutch government stated that the shortage of housing had increased up to 279,000 houses in 2021 and would continue to increase. Other reasons include an increase in single-person households, a shortage of construction land, high costs of construction, labour shortages, and nitrogen crisis-related issues. In order to combat the issue, the Dutch government intends to construct 900,000 new houses by 2030, where two-thirds of these will be rented and owner-occupied houses (Ministerie van Algemene Zaken, 2023b). Because housing is a basic human need, understanding the causes and consequences of this issue is essential. This research is supported by Housing market statistics dataset (CBS, 86098NED), Housing market price and development indicators (CBS, 82900NED), and Housing shortage and related indicators (CBS, 86065NED).

Data Sourcing

Description and limitations

1. huishouden_per_regio.csv (CBS dataset 71486NED)

The data are collected and published by Statistics Netherlands (CBS). CBS combines this information with other government administrative sources to determine how many households exist and what their composition is (for example: single-person households, couples, or families). The dataset covers annual measurements from 2000 onwards. The analysis of housing demand requires an understanding of the number of households and how that number changes over time. Ideally, each household requires its own housing unit, so changes in household structure (for example, more single-person households) have a strong impact on pressure on the housing market. Thus, this dataset provides a much-needed demand-side perspective to complement housing supply data. There are some limitations. Households are defined statistically on the basis of registration addresses, which may not always reflect real living situations (for example: students sharing housing or informal living). Misclassification can occur in temporary or complex housing arrangements (e.g. people registered at a parental home but living elsewhere).

2. vergunde_nieuwbouw.csv (CBS dataset 86098NED)

The data are collected and published by Statistics Netherlands (CBS). CBS processes this continuously updated administrative data to measure the life cycle of dwellings, including events such as new construction, demolition, additions, changes in use, and other housing stock movements. This dataset was selected because it provides insight into how the housing stock in the Netherlands changes over time, rather than only showing static numbers. The analysis filters and focuses on "vergunde nieuwbouw" to examine the change in newly started building projects after the 2020 Woningbouw Impuls. The dataset covers monthly data from January

2015 onwards. Although this dataset contains highly relevant information, it does not distinguish between types of demand (for example, student housing, family housing, or elderly housing), which limits its ability to show whether new developments match the areas where shortages are most severe. Even if the number of dwellings increases, it remains unclear which type of dwelling is expanding.

3. woningvoorraad.csv (CBS dataset 82900NED)

The database used is that of Statistics Netherlands, which collects and publishes the data from the records maintained in the Basisregistratie Adressen en Gebouwen (BAG). CBS publishes information about the housing stock annually. This includes data about ownership of houses, landlords, occupancy, and region-wise housing stock. This database was selected because it provides an indication of the amount of housing stock available in the Netherlands. It makes it possible to analyze the extent to which the number of dwelling units is rising in accordance with increased demand for housing. The analysis focuses on the total housing stock ("Totaal woningen") to determine whether the increase in the number of dwellings is able to meet housing demand. This database includes annual data starting in 2012 and covers all regions in the Netherlands. Despite the fact that the database provides valuable information about the total number of dwellings available, it does not provide information about housing affordability, waiting periods, or whether the homes are suitable for certain demographics.

4. wv_inkomen.csv (Centraal Bureau voor de Statistiek, 2026)

The data are collected and published by Statistics Netherlands. CBS combines different sources to provide information on occupied dwellings, household characteristics (such as age), housing tenure, and income levels across different regions in the Netherlands. This dataset was selected because it provides insight into the relationship between household income and housing conditions, which is relevant for understanding who is most affected by the housing shortage. The analysis focuses on household income categories (as a subgroup) to examine how housing is distributed among different income groups and whether lower-income households may face greater challenges in accessing suitable housing. The dataset contains annual data and covers all regions of the Netherlands. This dataset has its limitations, because it does not include income changes. Rather, income is measured at a specific point in time (an annual snapshot), which does not reflect short-term fluctuations or changes in financial situations throughout the year. Therefore, it cannot fully explain the extent to which different income groups experience difficulties in finding adequate housing.

Quantifying

Data cleaning

Upon initial import, a formatting issue was identified: several datasets used semicolons (;) rather than commas as delimiters. If left unaddressed, this would have resulted in significant loading errors or the merging of distinct columns into an uninterpretable dataset. To ensure a properly structured dataset from the outset, the data import functions were explicitly configured to read semicolon-separated files.

After importing, it was observed that missing values were recorded as the text string "Inkomen onbekend" (Income unknown). These were systematically recoded to R's native NA format. This step was essential, because retaining text to represent missing data would distort descriptive statistics and preclude any numerical modeling.

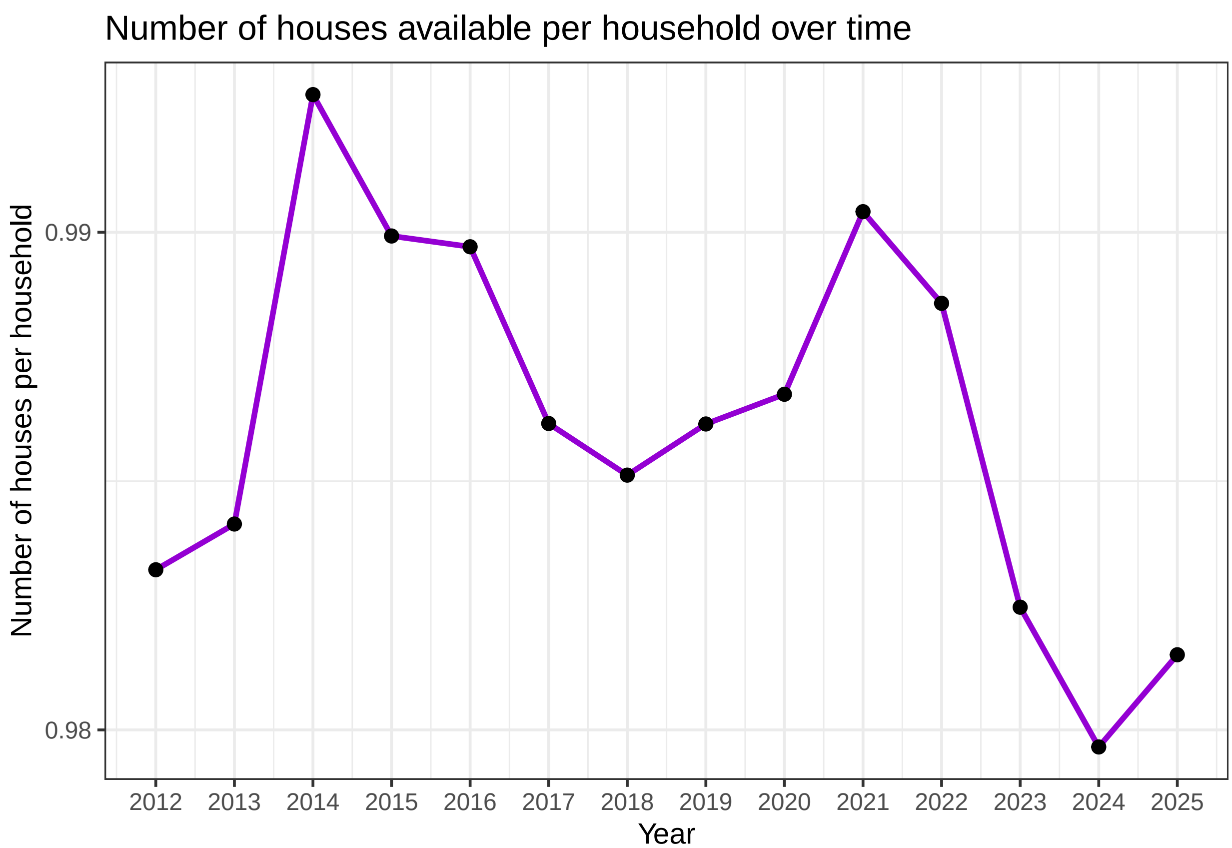
A further complication for the quantitative analysis was that several key columns, including total housing stock (totale.woningvoorraad..aantal), were read as character strings due to the presence of extraneous periods (.). To address this and enable mathematical operations, regular expressions were applied to remove these periods, after which the columns were converted to numeric variables using as.numeric(). This transformation was necessary for subsequent calculations of proportions, such as dividing household counts by population metrics.

Moreover, the raw datasets contained overlapping, aggregated categories within the Status.van.bewoning column. To prevent double-counting, the data were filtered to retain only the rows representing the actual population total (Totaal). The datasets were subsequently merged using a left join based on region (Regio.s) and time period (Perioden). Finally, to improve code clarity and prevent errors, the ambiguous raw column names were renamed to clear, logically descriptive labels.

Describing the variables

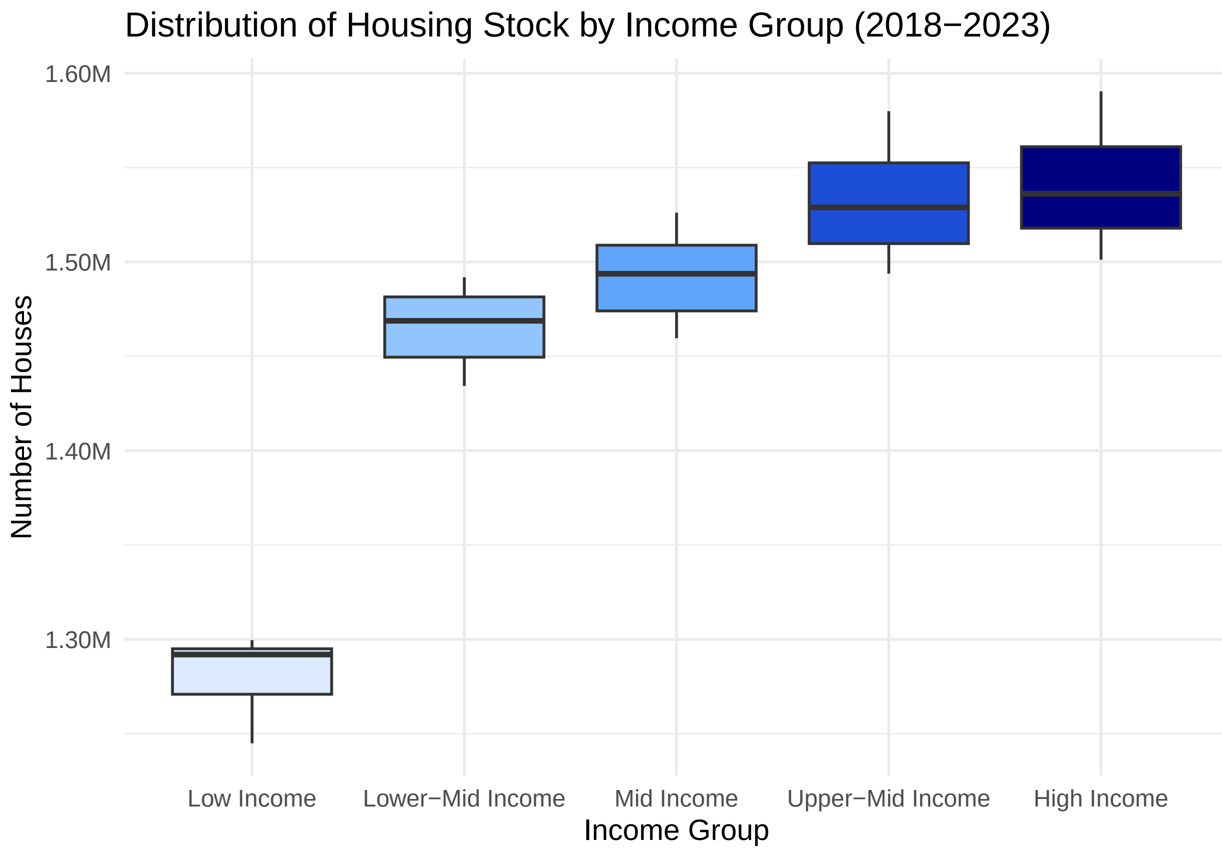
To enhance the analytical depth beyond descriptive summaries, two new variables were constructed. First, a Proportional Housing Share variable was created by dividing specific household groups by the cleaned total housing stock. This transformation converted absolute numbers into percentages, facilitating comparisons across regions irrespective of population size. Second, a Socio-Economic Structural Ratio was developed to contrast specific household dynamics against available housing capacity. This variable provides a more accurate assessment of local housing market pressures than the raw data alone.

Temporal variation



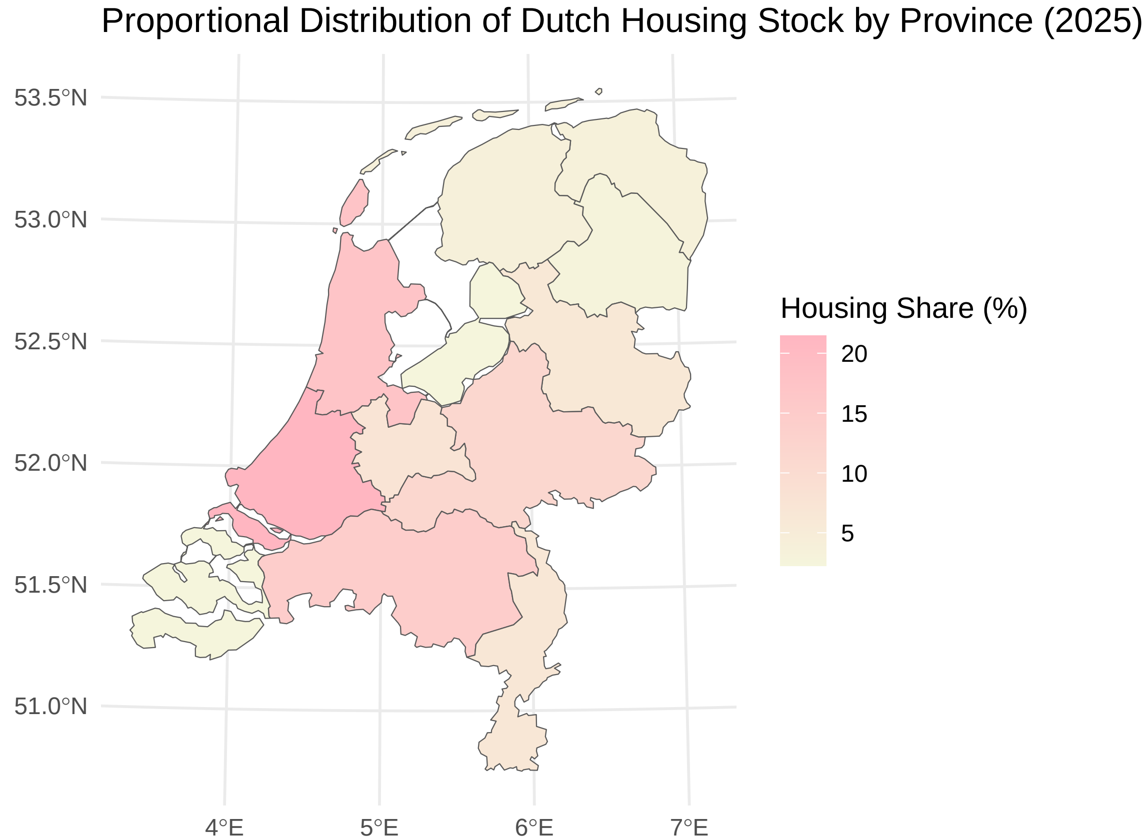
The timeline plot tracking housing availability per household between 2012 and 2025 reveals distinct cyclical fluctuations and a pronounced long-term decline. After a sharp peak in 2014, when availability reached its highest point at nearly 1.00 house per household, the ratio steadily decreased, reaching a notable low in 2018. Although a brief recovery occurred up to 2021, the market subsequently experienced a strong downward trend, reaching a low of fewer than 0.98 houses per household in 2024. These dynamics suggest that post-2021 macro-level factors, such as rising interest rates, supply chain disruptions, and elevated inflation, substantially constrained residential completions at a time when population growth was accelerating, thereby worsening the national housing shortage.

Sub-population variation



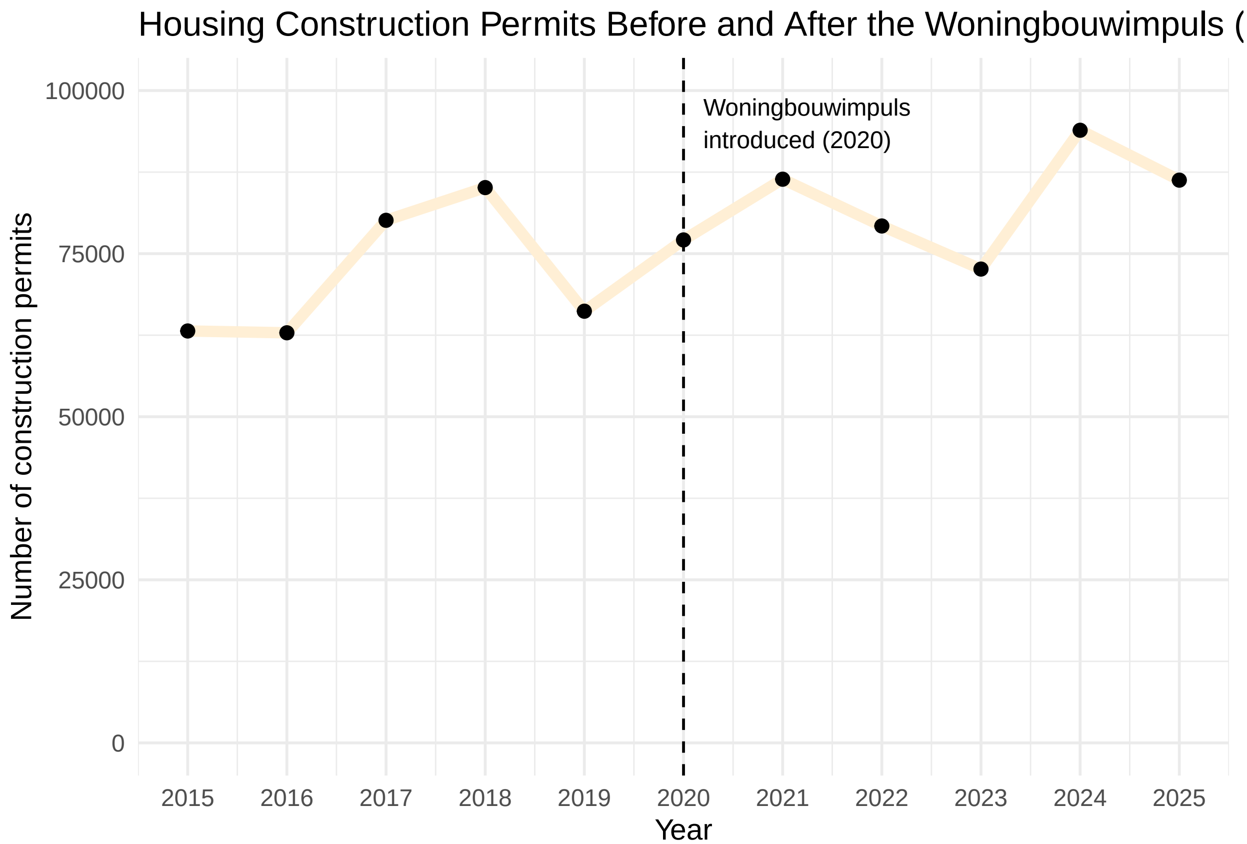
The boxplot analysis evaluates subgroup disparities and demonstrates a positive relationship between income levels and housing stock allocation across five distinct quintiles (20% income groups). The lowest income group, representing low-income households, accounts for a considerably smaller share of the housing stock, remaining below 1.3 million dwellings with limited variance over the five-year period. As income increases, the total volume of accessible housing rises markedly, with the highest income groups—representing high-income households—accounting for more than 1.5 million dwellings. The implications of these findings point to a pronounced socio-economic divide: the limited variance in the bottom quintile suggests a stagnant social and affordable housing sector that is not expanding, while higher-income market segments are characterized by significantly greater supply and mobility.

Spatial variation



The geographical distribution of the housing stock across the country is highly unequal. The spatial analysis highlights that the majority of residential properties are heavily concentrated in the western provinces, particularly South Holland and North Holland, both of which hold housing shares near or above 20%. In contrast, peripheral provinces such as Flevoland, Zeeland, and Drenthe exhibit considerably lower shares, often below 5%. This spatial clustering reflects the economic dominance of the core urban region. However, a closer examination of these mapping choices indicates that while absolute numbers show where the majority of the housing stock is located, they can obscure localized shortages in smaller, rapidly developing provinces such as Flevoland, where demand growth may exceed the supply implied by the low baseline stock share.

Event analysis



To assess the policy impact and its causal limitations, annual new construction permits were analyzed in relation to the introduction of the national Housing Construction Impulse program in 2020. Prior to the policy intervention, building permits were highly volatile, dropping significantly from a 2018 peak to approximately 66,000 in 2019. Following the 2020 implementation, permit approvals rose sharply, peaking at over 85,000 in 2021 and reaching a record high of nearly 93,000 in 2024. However, a careful causal assessment reveals the policy's limits: despite a substantial funding impulse, permits declined in 2023 and 2025. This indicates that while targeted government subsidies successfully stimulated project planning and development approvals, they cannot fully insulate the construction sector from broader macroeconomic shocks, such as fluctuating material costs and evolving nitrogen emission regulations.

Discussion

The empirical findings from the visualizations indicate that the Dutch housing market is characterised by deep structural problems. The spatial analysis presented in Section 1 shows that the housing stock is heavily concentrated in provinces such as South Holland and North Holland. This pattern corresponds with urban economics literature on geographic clustering, which argues that housing markets naturally follow regional job opportunities and urban centralising forces. However, as the analysis noted, focusing solely on absolute stock can obscure high housing pressures in smaller areas such as Flevoland. This observation aligns with academic critiques that broad spatial aggregates often mask localised supply shortages. Furthermore, the temporal timeline in Section 2 revealed a sharp decline in the number of houses available per household after 2021. This trend is extensively discussed in recent literature, which attributes the supply bottleneck to post-pandemic macro-level dynamics such as material price inflation and rising interest rates. Similarly, the sub-population analysis in Section 3 demonstrated that the lowest income group holds a small and static share of the housing stock over time. This fixed volume corresponds with housing studies that highlight a stagnant social housing sector. While higher-income groups experience flexible market expansion, lower-income households face severe structural barriers that limit their geographic mobility.

With regard to the policy event analysis in Section 4, a substantial increase in housing permits was observed following the 2020 Housing Construction Impulse policy, peaking at nearly 93,000 in 2024. Although it may appear straightforward to attribute this increase directly to the policy, such causal claims require caution. The policy was implemented alongside other major developments, including changes in local municipal zoning laws and fluctuating macroeconomic conditions. The visible declines in permits in 2023 and 2025 clearly demonstrate the limitations of policy intervention. A government subsidy programme can stimulate initial planning and funding, but it cannot fully insulate the construction market from external shocks such as high interest rates, supply chain disruptions, or environmental restrictions including the national nitrogen emissions crisis. Therefore, the data suggest a positive association between the policy and project planning, but do not support the claim that the policy alone resolves or controls the supply cycle.

Several limitations within the data pipeline should also be acknowledged. First, the dataset relies on annual snapshots, and this structural constraint prevents the observation of within-year adjustments, such as rapid seasonal real estate fluctuations or short-term shifts in household formation. Second, the raw data aggregate all housing types into a single total count and do not distinguish between student rooms, multi-family apartments, and single-family houses. Consequently, it cannot be determined whether the new permits correspond to the actual demographic needs of local populations. Third, the data cleaning process removed missing income records entirely. While this approach ensured a stable analytical environment, the absence of statistical imputation may have slightly reduced sample variation and marginally lowered model efficiency. To address these limitations, future research should incorporate quarterly or monthly tracking datasets to observe market responses to macro-level shocks in real time. Additionally, combining construction permit data with detailed demographic micro-data would allow researchers to examine whether new housing developments meet the specific structural needs of low-income and vulnerable groups.

Reproducibility

Github repository link

https://github.com/kysaad-source/PfE-groep-2.4-2

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